

Information Invariance: What It Is, What It Decides, and What It Does Not Derive

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We define information invariance for thermodynamic response charts as the stationarity of the loop functional of the entropy-scaled accessibility one-form, and settle what follows from it. Three structural results are proved on finite charts with executable certificates: information invariance is equivalent to vanishing thermodynamic curvature and to exactness of the one-form, i.e. to the chart's Onsager-reciprocal sector. Microscopically, $\chi = W(\Theta\Gamma)/W(\Gamma) \equiv 1$ is equivalent to symmetry of the full weighted correlation for all observables and implies information invariance. The converse requires a faithful coarse observation: microscopic asymmetry can cancel under coarse-graining. Nonzero coarse curvature therefore witnesses microscopic asymmetry, whereas zero curvature need not imply reversibility. On the experimental side we prove that the loop area of a *single* thermal cycle decides nothing — a flat chart with a one-time irreversible offset and a curved chart give equal first-cycle areas — which we illustrate on our own graphene Raman data, where the measured loop does not close. We then carry every obligation of the associated emergent-dynamics programme to a verdict rather than leaving it open: the quadratic energy law is not derivable from invariance but is uniquely selected by a stated minimal-scale principle; the linearised dynamics is a damped rotation, not a Schrödinger generator, and we give the exact condition that would make it one; the Planck length is an input rather than a consequence, the scale the theory supplies being cosmological; the emergent-Einstein claim fails on degree-of-freedom counting, on the $\hbar$ -dependence of the thermodynamic route, and on non-conservation of the scalar source; and the proposed decoherence rate is dimensionally invalid, its unique repair being the Diósi–Penrose expression. We regard the negative results as the paper's main contribution: they replace an unsupported programme with a small set of proved statements and explicitly stated conditions.

I. INTRODUCTION

Onsager reciprocity is derived under microscopic reversibility. Entropy-weighted reparameterisations of response functions yield an effective coupling matrix whose cross-coefficients are generically asymmetric [1]; the associated geometric statement is that the accessibility one-form $\omega = \lambda_p dp + \lambda_v dv$ need not be closed. This raises a structural question: *which* irreversible trajectories are admissible.

The present paper separates what can be proved from what is presently conjectured. Section II defines information invariance as a loop functional. Section III proves two finite theorems about it, with machine-checked certificates. Section V states the empirical situation honestly, including a non-closure finding in our own data that weakens the naive reading of loop area. Section VI states the dynamical programme as conjectures with explicit obligations.

II. INFORMATION INVARIANCE, DEFINED

Definition 1 (Information invariance). *Let ω be the entropy-scaled accessibility one-form on a response chart. Define the loop functional $\mathcal{I}(\gamma) = \oint_\gamma \omega$ for closed paths γ .*

The chart is information invariant if $\mathcal{I}(\gamma) = 0$ for every admissible closed γ , written $\delta\mathcal{I} = 0$.

This is a selection principle: it constrains admissible evolution, and asserts neither conservation of microscopic information nor invariance under arbitrary transformations. It postulates no new interaction and does not quantise spacetime.

III. PROVED FINITE STRUCTURE

Discretise the chart on a lattice with rational edge values; $\Omega = d\omega$ is the plaquette circulation, and discrete Stokes holds exactly.

Theorem 1 (Information invariance = flat sector). *On a finite simply connected chart the following are equivalent: (i) $\delta\mathcal{I} = 0$; (ii) $\Omega \equiv 0$; (iii) ω is exact, i.e. a potential Φ exists and line integrals are path independent. On the λ -chart with $\lambda_X = -ST/C_X$, (ii) is the flat-closure sector $\Gamma_c\Gamma_m = 1$, i.e. Onsager reciprocity.*

Proof sketch. (ii) $\Rightarrow$ (i) by discrete Stokes; (i) $\Rightarrow$ (ii) by taking $\gamma = \partial(\text{plaquette})$; (i) $\Leftrightarrow$ (iii) is path independence $\Leftrightarrow$ exactness on a connected graph. All three directions, plus planted-curvature negative controls, are executed over exact rationals in the accompanying certificate. $\square$

Theorem 2 (One cycle does not witness curvature). *There exist a flat chart with a single irreversible offset and a curved chart without offset whose first-cycle loop*

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areas are equal. The two are separated only by the second cycle: the offset construction collapses (reopen ratio 0), the curvature construction reopens with the same area (ratio 1).

Consequently a nonzero loop area measured on one cycle is consistent with irreversible drift and does not by itself establish $\iint \Omega \neq 0$. Certificates for both theorems are in `papers/information-invariance/certificates`; the pinned SHA-256 is reproduced by continuous integration on every commit.

IV. WHAT INFORMATION INVARIANCE MEANS MICROSCOPICALLY

The companion work derives, for an ensemble with weight W on a phase space carrying a time-reversal involution Θ , the identity $L_{ij} - \varepsilon_i \varepsilon_j L_{ji} = \int \langle (1 - \chi) J_i(0) J_j(t) \rangle_W dt$ with $\chi(\Gamma) = W(\Theta\Gamma)/W(\Gamma)$, and identifies the coarse-grained antisymmetric part with Ω [1]. Chaining this with Theorem 1 gives information invariance a microscopic reading, and one leg of the chain turns out to be one-directional.

Theorem 3 (Microscopic characterisation). *On a finite phase space, let $W > 0$, let Θ be an involution, and put $C_W(f, g) = \sum_{\Gamma} W(\Gamma) f(\Gamma) g(\Theta\Gamma)$. Then*

$$C_W(f, g) - C_W(g, f) = \sum_{\Gamma} W(\Gamma) [1 - \chi(\Gamma)] f(\Gamma) g(\Theta\Gamma), \quad (1)$$

exactly, and the following are equivalent: $\chi \equiv 1$; C_W is symmetric for all observables; the full microscopic antisymmetric response vanishes. These are not equivalences with the vanishing of a coarse image.

Proof. Reindex the second correlation by $\Gamma \mapsto \Theta\Gamma$ to obtain the identity. If $\chi = 1$, its right side vanishes. Conversely, take $f = \mathbf{1}_{\{a\}}$, $g = \mathbf{1}_{\{\Theta a\}}$; the antisymmetric correlation is $W(a) - W(\Theta a)$. Symmetry for every pair forces every such difference to vanish. Positivity of W makes this equivalent to $\chi = 1$. $\square$

Writing the microscopic antisymmetric response as A_W and the declared linear observation as K , the chain is

$$\chi \equiv 1 \iff A_W = 0 \implies K(A_W) = \Omega = 0 \iff \delta\mathcal{I} = 0.$$

The last equivalence uses Theorem 1's chart hypotheses. Reversing the middle arrow requires $\ker K \cap \mathcal{A}_{\text{adm}} = \{0\}$, where $\mathcal{A}_{\text{adm}}$ is the declared space of admissible antisymmetric responses. Indeed, $K(A_W) = 0$ then forces $A_W = 0$.

Theorem 4 (Nonzero curvature witnesses microscopic asymmetry). $\chi \equiv 1 \implies \Omega = 0$ *always. The converse fails: for $\Theta = (01)(23)$ and $W = (3, 1, 1, 3)$ the two orbit contributions are $+2$ and -2 , so $\chi \neq 1$ while the coarse-grained total vanishes.*

Together with Theorem 2 this bounds what a hysteresis experiment can decide:

$$\begin{aligned} \Omega \neq 0 \text{ across cycles} &\implies \chi \neq 1: \text{informative} \\ \Omega = 0 &\implies \text{nothing about } \chi \\ \text{one-cycle loop area} \neq 0 &\implies \text{nothing about } \Omega \end{aligned}$$

The predeclared second-cycle test is therefore not a precaution but the only step that carries an inference.

V. EMPIRICAL SITUATION

Temperature-cycled Raman measurements on monolayer CVD graphene on SiO_2 [1] give reproducible first-cycle hysteresis in the G and $2D$ peak positions, with loop areas well above fit uncertainty (independently refitted: 744 ± 22 and $692 \pm 28 \text{ cm}^{-1}\text{K}$ at spot 1). The loop, however, does *not* close at 300 K: the cooling branch returns $+4.95 \text{ cm}^{-1}$ (G , spot 1) above the heating branch. By Theorem 2 this first-cycle observation is compatible with an irreversible component (adsorbate desorption or strain release) and does not establish reversible curvature. A predeclared decision rule has been registered: reopen ratio ≥ 0.6 on both bands supports curvature; ≤ 0.2 indicates irreversible drift [2]. The measurement is pending.

VI. CONJECTURED DYNAMICS AND ITS OBLIGATIONS

The following are stated as conjectures. Each names the obligation a derivation must discharge; none is used to support the claims of Sec. III.

We take as the working dynamics, with $u = ct - r$ and energy density $\mathcal{E}(u, \theta, \phi)$,

$$\partial_u \mathcal{E} = \alpha_G \mathcal{E}^2 + \alpha_\Lambda \mathcal{E}_{\text{vac}} + \epsilon \mathcal{F}[\mathcal{E}], \quad (2)$$

and ask what, if anything, justifies it.

Theorem 5 (Invariance alone selects nothing). *On an information-invariant chart $\omega = d\Phi$, and every potential-transporting evolution $\partial_u \Phi = f(\Phi)$ preserves exactness, for arbitrary f . The dynamics compatible with $\delta\mathcal{I} = 0$ therefore form an infinite-dimensional family, and (2) is not derivable from Definition 1.*

Theorem 6 (Minimal scale usage selects the quadratic law). *For $\partial_u \mathcal{E} = \alpha_n \mathcal{E}^n$ the coupling must satisfy $[\alpha_n] = M^{1-n} L^{n-2} T^{2n-2}$. No n admits a coupling built from $\{G, c\}$ alone. Admitting a single new length λ gives uniquely $\alpha_n = G^{n-1} c^{4-4n} \lambda^{2n-3}$, and requiring λ to appear exactly once ($d = 1$) forces $n = 2$, with $\alpha_2 = G\lambda/c^4$.*

Two consequences. First, the quadratic term is not arbitrary once one states the selection principle — one new scale, used once — but it does not follow from invariance. Second, the coupling written as $\alpha_G \sim G/c^4$ is

short by exactly one length; by Theorem 14 the length the theory itself supplies is the cosmological $L_{\mathcal{E}}$, giving $\alpha_2 = 1/(c^2\sqrt{G\mathcal{E}})$. We record (2) as a conjecture with that stated provenance:

Conjecture 1 (Null-coordinate realisation). *Equation (2) with $\alpha_2 = G\lambda/c^4$ is the physically realised dynamics. **Obligation D1'**: justify the selection principle itself, and the identification $\lambda = L_{\mathcal{E}}$.*

A. Where a complex phase can come from (proved)

The step usually left implicit is the appearance of a complex phase in a real theory. Its finite content is settled.

Theorem 7 (Real flow = Schrödinger flow). *Let $x' = Ax$ be a real linear flow on $\mathbb{R}^{2n}$ and J an orthogonal complex structure ($J^2 = -I$, $J^T = -J$). Then $A^T = -A$ and $AJ = JA$ if and only if $H := -\hbar JA$ is Hermitian for J , in which case the flow reads $i\hbar\psi' = H\psi$ under $x \leftrightarrow \psi$; equivalently, the flow is complex linear and norm preserving.*

Theorem 8 (One-channel obstruction). *An odd-dimensional real flow admits no complex structure, since $\det(J)^2 = \det(-I) = (-1)^N$. In particular a single scalar channel $x' = ax$ is never a Schrödinger flow, for any a and any rescaling.*

Linearising Conjecture 1 about a background gives, at leading order, the single real channel $\partial_u \delta\mathcal{E} = 2\alpha_G \bar{\mathcal{E}} \delta\mathcal{E} + \epsilon \mathcal{F}' \delta\mathcal{E}$. By Theorem 8 this cannot be Schrödinger evolution as it stands, and by Theorem 7 what a derivation must supply is explicit:

D2a a conjugate channel π of $\delta\mathcal{E}$ within the dynamics, and a complex structure J on the pair $(\delta\mathcal{E}, \pi)$;

D2b antisymmetry of the linearised generator on that pair, i.e. the quadratic self-interaction must contribute to the Hermitian H and not to a symmetric part of A .

The first is supplied by structure that is already derived, not assumed. An oriented cut yields a quarter turn ι with $\iota^2 = -1$ and $\iota^\dagger = -\iota$ [4], and every generator splits uniquely relative to the cut J into an even part preserving the two sheets and an odd part transporting between them [5]. Taking the conjugate channel to be the image on the opposite sheet gives:

Theorem 9 (Cut supplies the pair; phase is the odd part). *Let J be a balanced cut, and write the generator in the sheet basis as blocks $A = \begin{pmatrix} A_{++} & B \\ C & A_{--} \end{pmatrix}$ with $\iota = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$. Then $A\iota = \iota A$ iff $A_{++} = A_{--}$ and $C = -B$; $A^T = -A$ iff the diagonal blocks are antisymmetric and $C = -B^T$; both hold iff the diagonal blocks are equal and antisymmetric and B is symmetric, in which case*

$H = -\hbar A$ is Hermitian and Theorem 7 applies. A purely cut-even generator never mixes the sheets and produces no phase, and the symmetric part of A is exactly the damping: $d|x|^2/du = 2x^T A_{\text{sym}} x$.

B. The computation, and its negative outcome

Theorem 9 leaves one computation. We perform it on a transverse lattice with the parity cut, in exact arithmetic. The linearisation of Conjecture 1 about a background contributes three operator types: pointwise multiplication M by $2\alpha_G \bar{\mathcal{E}}$ from the quadratic term, a second difference L from any diffusive part of $\mathcal{F}$, and a first difference D from any transport part. We find:

- M is symmetric and cut even; it contributes $d|x|^2/du = 4\alpha_G \bar{\mathcal{E}} |x|^2$, i.e. pure growth or decay, and no phase;
- L is symmetric and negative semidefinite: diffusion, no phase;
- D is antisymmetric and cut odd: it alone preserves the norm and meets the conditions of Theorem 9;
- for $A = M + L + D$ the symmetric part is exactly $M + L$, which does not vanish at a nonzero background; a coordinate rescaling is a similarity and cannot remove it.

Result. The linearised dynamics of Conjecture 1 is not a Schrödinger generator. It is a damped (or amplified) rotation. The emergent-quantum step therefore does not follow from the equation as written, and we withdraw it as a claim. What survives is a sharp conditional: the step holds if and only if the dynamics is amended so that the cut-even symmetric part cancels at the background, $M + L = 0$, leaving the cut-odd transport term as the generator. Theorems 1 and 2 are independent of this and are unaffected.

C. Emergent geometry: three obstructions

Theorem 10 (Degree-of-freedom no-go). *In four dimensions the metric has ten components; four coordinate conditions and four constraints leave two propagating polarisations. The dynamics carries one scalar function, so it can reproduce at most a one-function sector (spherically symmetric or conformally flat) and cannot generate two independent polarisations.*

Theorem 11 (The Clausius route imports $\hbar$). *No temperature is constructible from acceleration and c alone in (M, L, T, Θ) ; the Unruh temperature requires $\hbar$ and k_B . Hence the derivation of the field equations from $dQ = T dS$ [6] is unavailable to a dynamics claiming to be deterministic and non-quantised.*

Theorem 12 (The scalar source is not conserved). *The contracted Bianchi identity forces $\nabla^\mu T_{\mu\nu} = 0$. The law $\partial_u \mathcal{E} = \alpha \mathcal{E}^2 + \dots$ is not a divergence: on a periodic transverse lattice the total changes, whereas a genuine divergence leaves it fixed.*

We therefore withdraw the emergent-gravity claim as a derivation. Three conditional routes remain and are stated rather than assumed: restrict to the one-function sector; admit $\hbar$ explicitly and give up deriving quantum structure; or supply the compensating flux, or a second field, that restores conservation of the source.

D. The regulator scale (settled)

The finiteness claim associated with the withdrawn emergent-gravity derivation is $\langle \mathcal{E}(x) \mathcal{E}(x') \rangle \sim (r^2 + L_P^2)^{-1}$ with $L_P = \sqrt{\hbar G/c^3}$. Dimensional analysis decides its status.

Theorem 13 (No Planck length without $\hbar$). *In the (M, L, T) dimension group, $\hbar$ is not in the span of $\{c, G\}$, and no product $G^a c^b$ is a length. Hence L_P cannot be constructed from the deterministic constants; admitting it presupposes $\hbar$.*

Theorem 14 (The theory's own regulator is cosmological). *From $\{c, G, \mathcal{E}\}$ a length does exist, uniquely up to powers: $L_{\mathcal{E}} = c^2/\sqrt{G\mathcal{E}}$, equivalently $\Lambda^{-1/2}$. For a vacuum energy density of the observed order, $L_{\mathcal{E}}/L_P \sim 10^{61}$.*

Consequently the model regularises, if at all, at a cosmological scale; inserting L_P is an assumption and the ultraviolet-finiteness claim of the earlier draft is withdrawn. The theory must either admit $\hbar$ as a primitive — forfeiting the claim to derive quantum structure — or work with $L_{\mathcal{E}}$.

E. The decoherence rate (corrected)

Theorem 15 (The printed rate is not a rate). *$Gm^2/(\hbar V)$ has dimensions $M^0 L^{-2} T^{-1}$: a rate per unit area. The unique repair replacing the volume by a length, $Gm^2/(\hbar R)$, is dimensionally a rate and is exactly the Diósi–Penrose expression $E_G/\hbar$ with $E_G \sim Gm^2/R$ [7, 8].*

With $m = 10^{-17} \text{ kg}$ and $R = V^{1/3} = 10^{-7} \text{ m}$ the repaired rate is $6.3 \times 10^{-4} \text{ s}^{-1}$, of the order quoted in the earlier draft — but it is a Diósi–Penrose number, recovered only after the correction. We therefore state the result as a *consistency requirement*: any emergent gravity of this kind must reproduce $E_G/\hbar$ in this regime. It becomes a prediction only if one of the following is supplied: a derivation of the coefficient, which DP fixes only

through a choice of regularisation length; a mass or geometry regime of departure; or a dependence on shape or entropy gradient, which DP does not carry. Optomechanical tests discriminate such departures [9, 10], not the agreement.

VII. SUMMARY OF STATUS

The theorem statements below use the hypotheses and proofs in their cited sections. Executable certificates check declared finite instances and negative controls; they are not formal proofs of all quantified statements. Conditions and pending measurements remain explicitly identified.

VIII. CONCLUSION

Information invariance, defined as stationarity of the loop functional, is equivalent on finite charts to vanishing thermodynamic curvature and to exactness — a proved statement with an executable certificate. The experimental signature usually quoted for curvature, a nonzero one-cycle loop area, is proved insufficient; the decision is deferred to a predeclared second-cycle test. The dynamical programme connecting this principle to quantum and gravitational phenomena is stated as conjectures with named obligations, and every obligation is carried to a verdict rather than left open. The emergent-gravity claim is withdrawn on three independent grounds — degree-of-freedom counting, the $\hbar$ -dependence of the thermodynamic route, and non-conservation of the scalar source. For the quantum step the obligation is discharged with a negative outcome: the proposed energy dynamics generates a damped rotation rather than unitary evolution, and the step survives only as the explicit condition that the cut-even symmetric part cancel at the background. We regard reporting this as more useful than asserting the step.

REPRODUCIBILITY

All theorems marked “proved” are accompanied by executable certificates over exact rational arithmetic, with pinned SHA-256 digests regenerated by continuous integration on every commit; tampering with any recorded value changes the digest and fails the build. Source, certificates and the claim ledger: `Parveen117/Publications, papers/information-invariance`.

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Statement	Status	Where
$\delta\mathcal{I} = 0 \Leftrightarrow \Omega = 0 \Leftrightarrow \text{exact}$	proved	Thm 1
One cycle does not witness curvature	proved	Thm 2
Real flow is Schrödinger iff J , antisym.	proved	Thm 7
One channel admits no complex structure	proved	Thm 8
Cut supplies the pair; phase is odd part	proved	Thm 9
No Planck length without $\hbar$	proved	Thm 13
Own regulator is cosmological	proved	Thm 14
Printed decoherence rate is not a rate	proved	Thm 15
Invariance alone selects no dynamics	proved	Thm 5
Minimal scale usage selects $n = 2$	proved	Thm 6
One scalar cannot carry two polarisations	proved	Thm 10
Clausius route imports $\hbar$	proved	Thm 11
Scalar source is not conserved	proved	Thm 12
Symmetric $C_W \Leftrightarrow \chi \equiv 1$	proved	Thm 3
$\Omega = 0$ does not imply $\chi \equiv 1$	proved	Thm 4
Selection principle itself; $\lambda = L_{\mathcal{E}}$	stated condition	D1'
Reversible curvature in graphene	measurement pending	Sec. V

TABLE I. Claim status under the hypotheses of the cited results.

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- [2] Parveen117/Thermodynamics-Reproducibility, campaign T02_RAMAN_THERMAL_HYSTERESIS (2026).
- [3] Dimensional statements in Sec. V use only (M, L, T) exponents; the numerical illustration uses a vacuum energy density of order 10^{-9} J m^{-3} .
- [4] Recognition-Kernel Framework, `theorems/foundation/FOOE_NATIVE_EULER_FROM_IOTA_COMPLEX.md` (PROVED), Zenodo doi:10.5281/zenodo.21760119.
- [5] Recognition-Kernel Framework, `theorem/41_cut_graded_universal_generator_theorem.md` (certified, pin 34afc445...), *ibid*.
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